# LAB EXPERIMENT RECORD

## Set 19: Survey Responses Analysis

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|---|---|
| **Department** | AIDS |
| **Course Code / Course Name** | DSA0602/ Data Handling and Visualization |
| **Experiment** | Data Visualization using R |
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| **Register Number** | 192424049 |
| **Date** | 13/10/2026 |
| **Day** | Tuesday |

## AIM

To analyze survey responses using Tableau by creating a grouped bar chart to visualize the distribution of answers for Question 1, a stacked bar chart to represent the overall distribution of responses across all three questions, a table to display survey response details, and an interactive dashboard combining all visualizations.

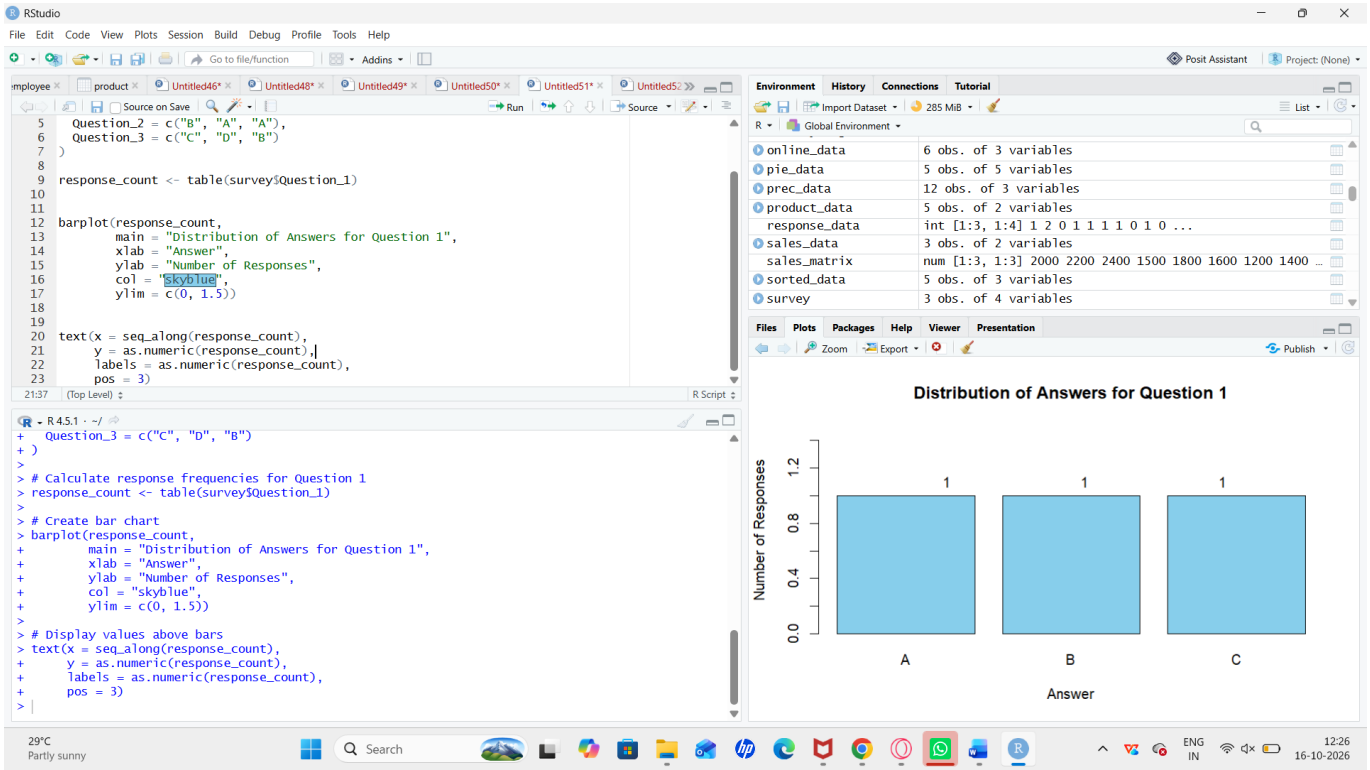
1. Create a grouped bar chart to visualize the distribution of answers for Question

```
survey <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

response_count <- table(survey$Question_1)


barplot(response_count,
        main = "Distribution of Answers for Question 1",
        xlab = "Answer",
        ylab = "Number of Responses",
        col = "skyblue",
        ylim = c(0, 1.5))


text(x = seq_along(response_count),
     y = as.numeric(response_count),
     labels = as.numeric(response_count),
     pos = 3)
```

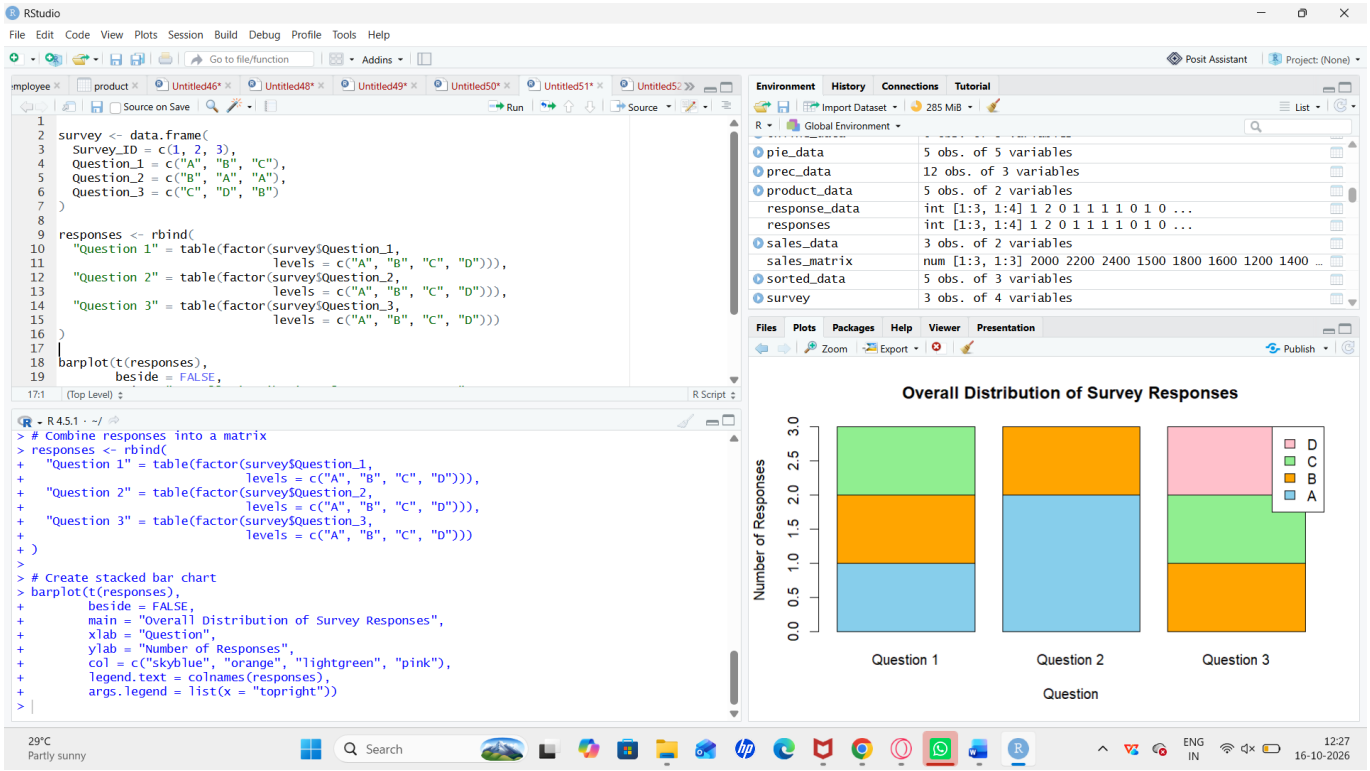


2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

```
survey <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

responses <- rbind(
  "Question 1" = table(factor(survey$Question_1,
                              levels = c("A", "B", "C", "D"))),
  "Question 2" = table(factor(survey$Question_2,
                              levels = c("A", "B", "C", "D"))),
  "Question 3" = table(factor(survey$Question_3,
                              levels = c("A", "B", "C", "D")))
)

barplot(t(responses),
        beside = FALSE,
        main = "Overall Distribution of Survey Responses",
        xlab = "Question",
        ylab = "Number of Responses",
        col = c("skyblue", "orange", "lightgreen", "pink"),
        legend.text = colnames(responses),
        args.legend = list(x = "topright"))
```

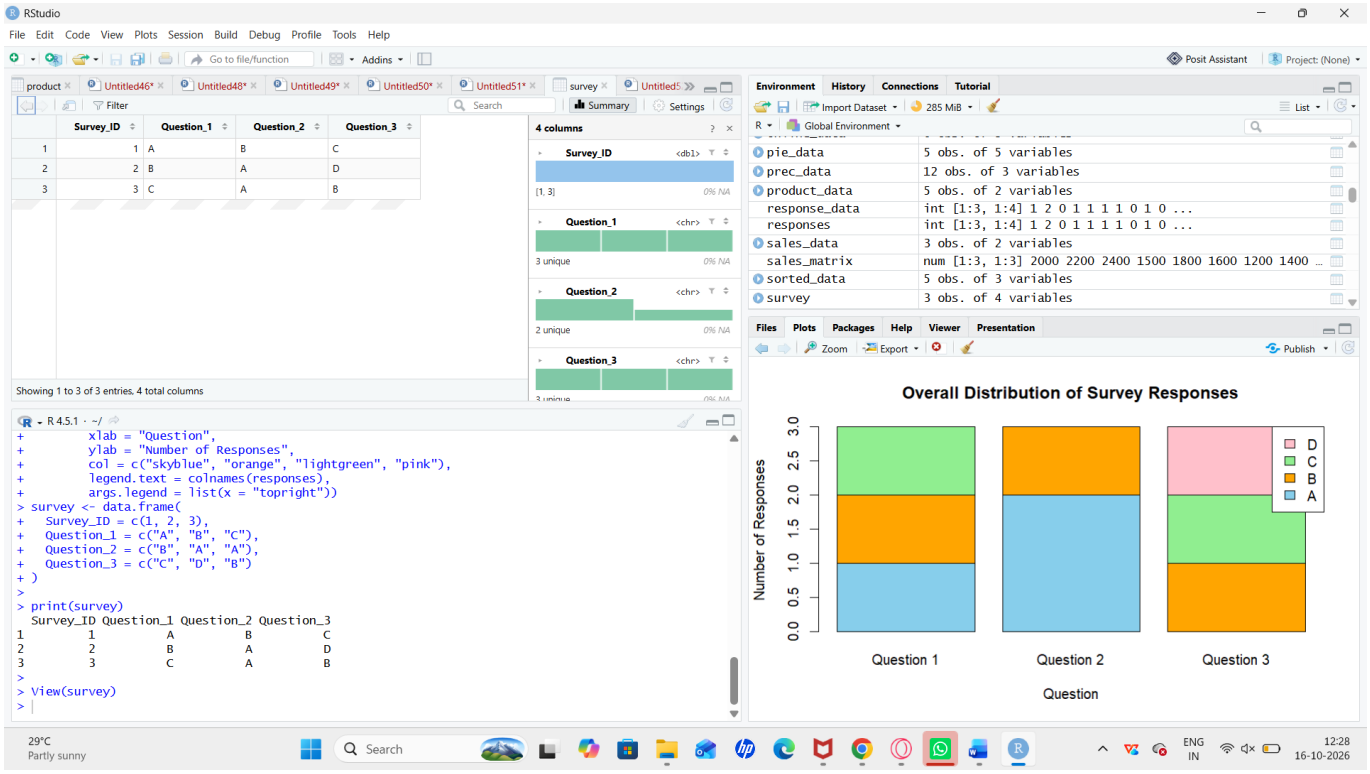


3.Build a table to show the survey response data for each survey.

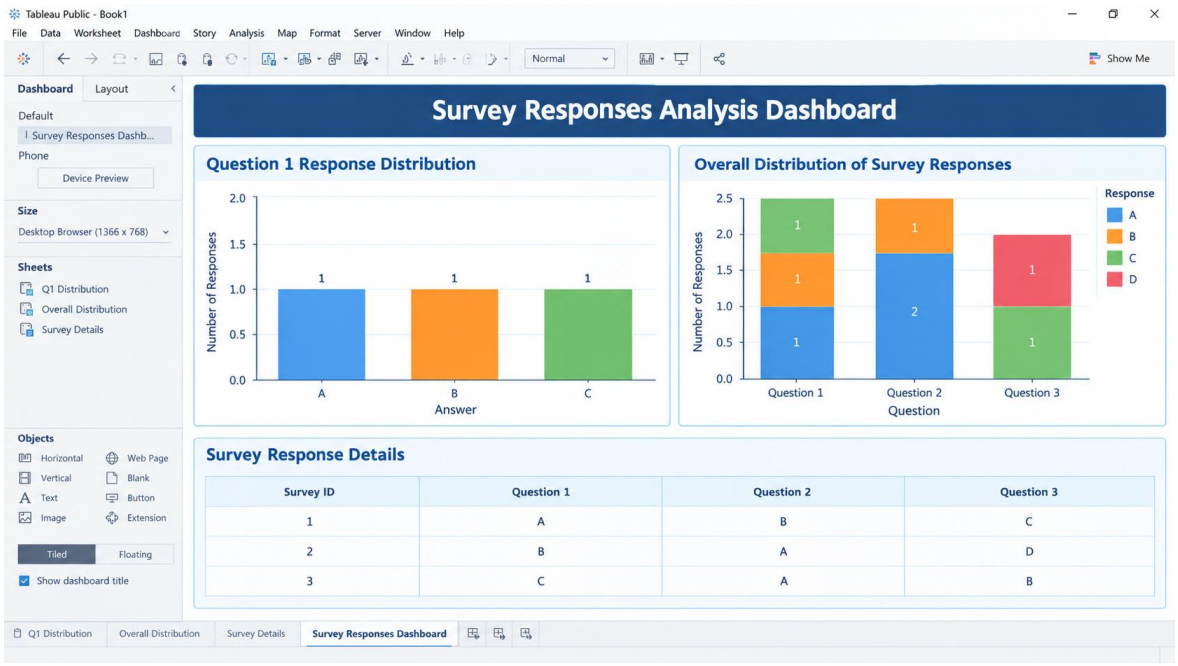
```
survey <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

print(survey)

View(survey)
```



**4**.Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive



Procedure

1. Prepare the survey dataset containing Survey ID, Question 1, Question 2, and Question 3.

2. Open Tableau and connect the survey dataset using the Text File option.

3. Create a grouped bar chart by placing the Question 1 response categories on Columns and the count of Survey ID on Rows. Add appropriate labels and a chart title.

4. Create a stacked bar chart to represent responses across all three questions. Pivot Question 1, Question 2, and Question 3 into a single Question field and a Response field, then use the question and response fields to construct the chart.

5. Create a text table displaying Survey ID and the responses to Questions 1, 2, and 3. Add a suitable title.

6. Create a new dashboard and combine the grouped bar chart, stacked bar chart, and survey response table.

7. Adjust the layout, display the dashboard title, and save the Tableau workbook.

Result

Thus, the survey response data was successfully analyzed using Tableau. A grouped bar chart was created to display the distribution of answers for Question 1, a stacked bar chart was used to represent responses across all three questions, and a table was created to display individual survey responses. Finally, all three visualizations were combined into an interactive dashboard for effective survey response analysis.